

Assignment Answers

Foundations, Lifecycle, and Real-World Applications

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Topics covered: Data Science & ML Fundamentals, Lifecycle, Supervised vs. Unsupervised Learning, Overfitting, Train/Test Splits, Case Study

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1 Defining Data Science and Machine Learning

Companies and researchers collect huge amounts of data: clicks, transactions, sensor readings, medical records. None of it is useful by itself. **Data science** is the discipline that turns raw data into decisions, predictions, and products.

Machine learning is a branch of artificial intelligence that learns patterns directly from data, instead of following rules a human writes out by hand.

The relationship: Machine learning sits inside data science. It is not a separate field running next to it. Data science is the bigger umbrella, covering the whole path from raw data to a useful answer: collecting it, cleaning it, exploring it, modeling it, and communicating what came out of it. Machine learning is specifically the modeling step, the part where something learns to predict or find patterns on its own.

So every machine learning project counts as a data science project, but plenty of data science work never touches machine learning at all.

REAL-LIFE EXAMPLE

An online clothing retailer, call it **Hashim Tailor**, notices revenue has been flat for two quarters. Leadership suspects customers are leaving quietly — no complaints, no canceled accounts, just people who stop buying. They bring in a data team to find out who is about to leave before it happens. Churn prediction works well as an example because nearly every stage of data science shows up here, with machine learning as just one piece of it.

2 The Data Science Lifecycle

The lifecycle runs from a vague business question to a working solution in production: defining the problem, collecting data, cleaning and preprocessing it, exploring it, modeling it, evaluating the model, then deploying it.

Stage	What Happens
1. Problem Definition	Turn a vague business worry into a measurable question
2. Data Collection	Pull data from relevant systems and sources
3. Cleaning & Preprocessing	Fix missing values, inconsistencies, and scale features
4. Exploratory Data Analysis	Visualize data to find patterns before modeling
5. Modeling (<i>Machine Learning</i>)	Train an algorithm to learn patterns and make predictions
6. Evaluation	Test the model on unseen data and measure performance
7. Deployment	Put the model into production where it drives real decisions

Where Machine Learning fits: ML fits specifically at the modeling stage — one step out of seven, but the centerpiece for problems like churn prediction. It fits there because the problem depends on subtle, interacting signals that are hard to hand-code as rules, so the model learns the pattern from historical examples instead of being told it directly. The other six stages are plain data science: defining the question, gathering data, cleaning it, exploring it, judging what the model produced, and turning that into something the business actually does.

3 Supervised vs. Unsupervised Learning

Supervised learning trains on labeled data, where the correct answer is already known, and predicts that answer for new cases.

EXAMPLE

Predicting whether an e-commerce customer will churn, using past customers already labeled "churned" or "stayed."

Unsupervised learning trains on unlabeled data — no answer given at all — and finds hidden patterns or groupings on its own.

EXAMPLE

Clustering e-commerce customers into segments, like "frequent small spenders" or "occasional big buyers," with no predefined categories.

	Supervised Learning	Unsupervised Learning
Data used	Labeled (answer known)	Unlabeled (no answer given)
Goal	Predict a known outcome	Discover hidden structure
E-commerce example	Predicting churn using customers labeled "churned" or "stayed"	Grouping customers into segments with no predefined categories
Output	A specific prediction	Clusters or patterns, interpreted afterward
How you check it worked	Compare predictions to actual outcomes	Judge whether groupings make practical sense

Key difference: supervised learning answers a question you already know how to check. Unsupervised learning reveals structure you didn't know to look for in the first place.

4 Overfitting: Causes & Prevention

Overfitting happens when a model learns the training data too well — noise and random quirks included — instead of the general pattern underneath it. It looks great on training data and falls apart on new data, because it memorized rather than understood.

Common Causes

- Too little training data
- A model too complex for the problem (too many parameters)
- Training for too long
- Too many irrelevant features

How to Prevent It

- **More data** — noise is harder to memorize when there's more variety in the examples
- **Simplify the model** — fewer parameters, less complexity
- **Cross-validation** — test across multiple data splits to catch overfitting early
- **Regularization** — penalize the model for excess complexity
- **Early stopping** — halt training once validation performance stalls
- **Train/test split** — always evaluate on data the model never saw while training

5 Training & Test Data Splits

The dataset gets divided into two parts, usually **70-80% for training** and **20-30% for testing**. The model learns only from the training portion. The test portion stays untouched until evaluation.

Why This Matters

- It is the only way to check whether the model learned a general pattern rather than memorizing the training set
- Testing on the same data used for training gives a falsely perfect score — overfitting would go unnoticed
- The split simulates how the model will actually behave on real, unseen data down the line

6 Case Study: Machine Learning in E-Commerce

Source: "E-Commerce Product Recommendation System based on ML Algorithms," research paper, arxiv.org/pdf/2407.21026

What the Study Did

Researchers built a system to generate personalized product recommendations for individual customers. They used PCA (Principal Component Analysis) to cut down the number of features, then tested four algorithms against each other — Gaussian Naive Bayes, Random Forest, Logistic Regression, and Decision Tree — to see which predicted customer preferences best.

Findings

Random Forest won clearly: **99.6% accuracy**, a **96.99 R-squared score**, **1.92% MSE**, and **0.087 MAE**. Its predictions landed extremely close to what customers actually wanted. The researchers read the result as a win on both sides — more relevant recommendations for customers, more engagement and sales for the business.

Random Forest works by building many decision trees and averaging their predictions, instead of relying on one tree or one straight line through the data — which goes a long way toward explaining why it beat the other three algorithms by this much.

Mapping to the Data Science Lifecycle

Stage	Covered?	What Happened
Problem Definition	Yes	Build recommendations that improve buying decisions
Data Collection	Implied	Customer browsing and purchase data feeds the model
Cleaning / Preprocessing	Yes	PCA reduces and simplifies features — the same role scaling plays elsewhere
Exploration	Barely	The paper focuses mainly on modeling
Modeling	Yes — core focus	Four algorithms trained and compared
Evaluation	Yes	Accuracy, R-squared, MSE, MAE used to judge the winner
Deployment	Not covered	The paper stops at building and testing the model

Most of the work in this paper sits at modeling and evaluation — exactly where machine learning lives inside the bigger data science pipeline. PCA does what scaling does elsewhere: makes messy data usable. The four-way comparison shows how data scientists actually pick a model — test several, let the numbers decide.